

Data Cleaning Using Constraints

Khalid Belhajjame

What is a Data Error?

Error = A value that is different from ground truth

What is a Data Error?

ID	FN	LN	ROLE	ZIP	ST	SAL
105	Anne	Nash	Emp	85281	NY	110
211	Mark	White	Man	15544	NY	80
386	Mark	Lee	M	85281	AZ	75
215	Anna	Smith Nash	Emp	85283		1

Constraint violation

Duplicates

Pattern Violation

Outliers

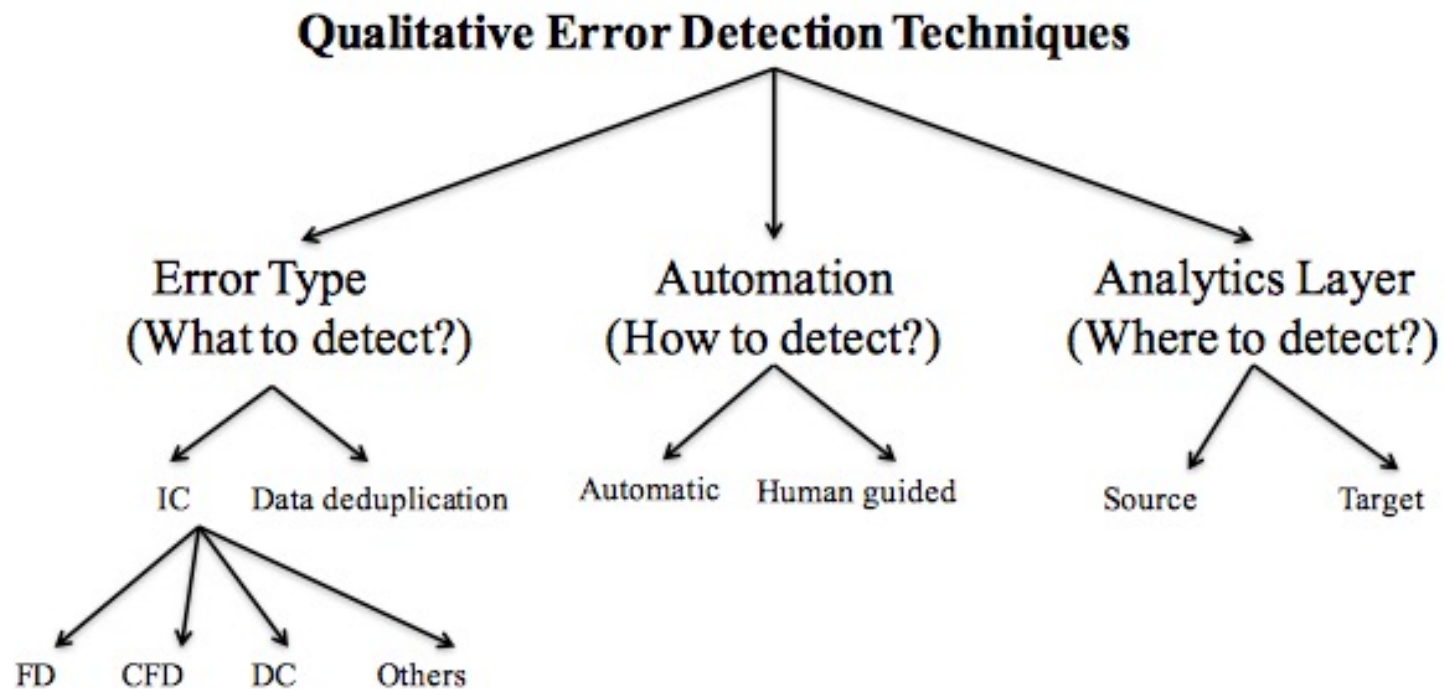
Error Detection Strategies

- Rule-based detection algorithms
 - Constraint violations, FDs, CFDs, Denial Constraints
- Pattern verification and enforcement
 - Syntactic patterns (date formatting)
 - Semantic patterns (location names WI)
- Quantitative methods
 - Statistical outliers
- Deduplication

Variety of tools

	DBoost	DC-Clean	OpenRefine	Trifacta	Pentaho	KNIME	Katara	Tamr
Pattern violations			✓	✓	✓	✓	✓	
Constraint violations		✓						
Outliers	✓							
Duplicates								✓

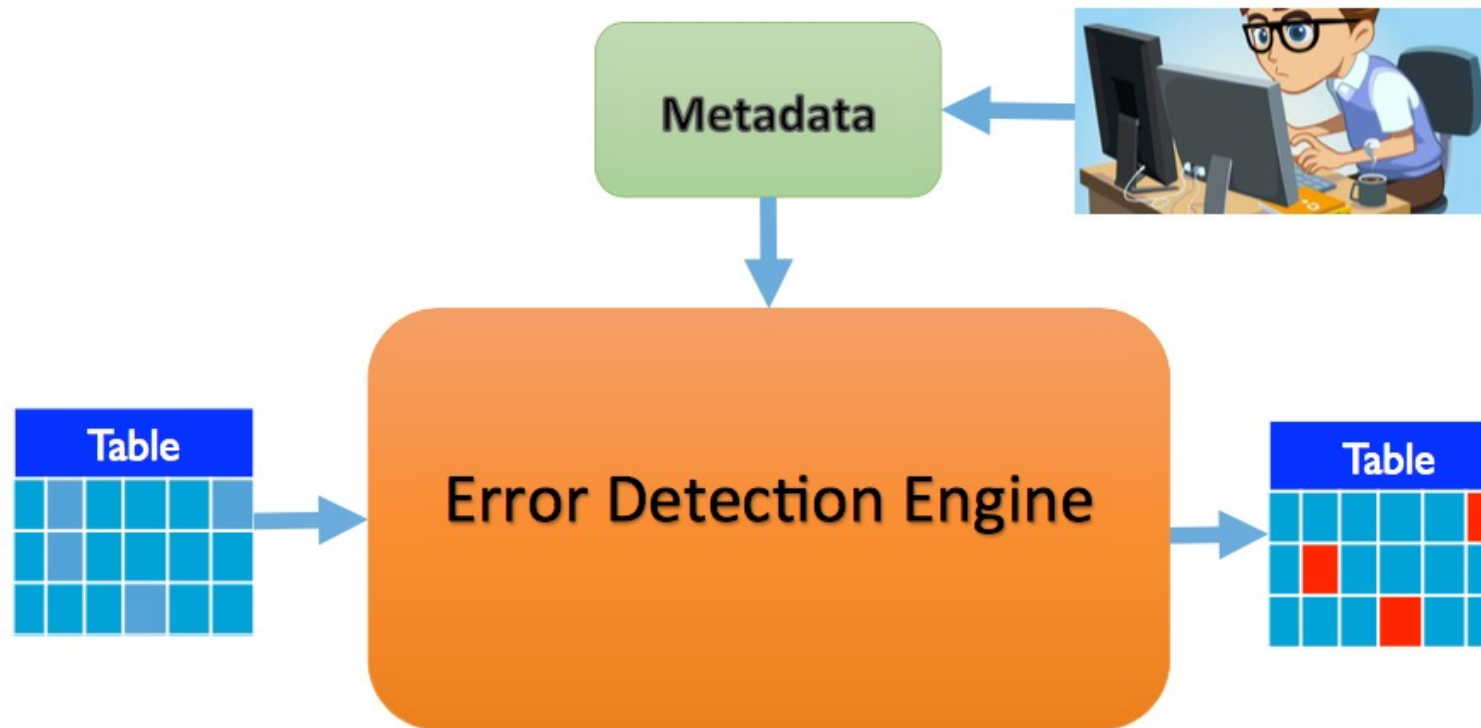
Error Detection Taxonomy



Business Intelligence Layer (Where to Detect?)

- Errors can happen in all stages of a business intelligence (BI) stack, for example, errors in the source database are often propagated through the data processing pipeline.
- While most anomaly detection techniques detect errors in the original database, some errors can only be discovered much later in the data processing pipeline, where more semantics and business logics becomes available, for example, constraints on total budget can only be enforced after aggregating cost and expenses.

Error detection engine



3. COMBINING ERROR DETECTORS

Lots of Detectors

Constraint violation
 $P = 1/4$

ID	FN	LN	ROLE	ZIP	ST	SAL
105	Anne	Nash	Emp	85281	NY	110
211	Mark	White	Man	15544	NY	80
386	Mark	Lee	M	85281	AZ	75
215	Anna	Smith Nash	Emp	85283		1

Pattern Violation
 $P = 1/1$

Error Detection Strategies

- Rule-based detection algorithms
 - Constraint violations, FDs, CFDs, Denial Constraints
 - Any subset of data that does not conform to the defined rules is considered erroneous, which is also referred to as a violation.

Integrity Constraints

- Functional Dependencies
- Conditional Functional Dependencies
- Denial Constraints

Functional Dependency

A **functional dependency** (FD) ϕ is defined as $X \rightarrow Y$, where $X \subseteq \text{attr}(R)$ and $Y \subseteq \text{attr}(R)$. An instance I of R satisfies FD ϕ , denoted as $I \models \phi$ if for any two tuples t_α, t_β in I , such that $t_\alpha[X] = t_\beta[X]$, then $t_\alpha[Y] = t_\beta[Y]$

Functional Dependency (Example)

- Consider the tax records in the following table.
- Each record describes an individual's address and tax information with: first and last name (FN, LN), gender (GD), area code (AC), mobile phone number (PH), city (CT), state (ST), zip code (ZIP), marital status (MS), has children (CH), salary (SAL), tax rate (TR), tax exemption amount if single (STX), married (MTX), and having children (CTX).

TID	FN	LN	GD	AC	PH	CT	ST	ZIP	MS	CH	SAL	TR	STX	MTX	CTX
t_1	Mark	Ballin	M	304	232-7667	Anthony	WA	25813	S	Y	70000	3	2000	0	2000
t_2	Chunho	Black	M	206	154-4816	Seattle	WA	98103	M	N	60000	4.63	0	0	0
t_3	Annja	Rebizant	F	636	604-2692	Cyrene	MO	64739	M	N	40000	6	0	4200	0
t_4	Annie	Puerta	F	501	378-7304	West Crossett	AR	72045	M	N	85000	7.22	0	40	0
t_5	Anthony	Landram	M	319	150-3642	Gifford	IA	52404	S	Y	15000	2.48	40	0	40
t_6	Mark	Murro	M	970	190-3324	Denver	CO	80251	S	Y	60000	4.63	0	0	0
t_7	Ruby	Billinghurst	F	501	154-4816	Kremlin	AR	72045	M	Y	70000	7	0	35	1000
t_8	Marcelino	Nuth	F	304	540-4707	Kyle	WV	25813	M	N	10000	4	0	0	0
t_9	Ann	Puerta	F		378-7304	West Crossett	AR	72045	M	N	86000	7.22	0	40	0

Functional Dependency (Example)

- The following FD: ZIP $\rightarrow$ ST, needs to hold.
- Does it?

TID	FN	LN	GD	AC	PH	CT	ST	ZIP	MS	CH	SAL	TR	STX	MTX	CTX
t_1	Mark	Ballin	M	304	232-7667	Anthony	WA	25813	S	Y	70000	3	2000	0	2000
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Functional Dependency (Example)

- The following FD: ZIP $\rightarrow$ ST, needs to hold.
- Does it?
- No, it doesn't. the tuples t_1 and t_8 associates different states to the same zip code.

TID	FN	LN	GD	AC	PH	CT	ST	ZIP	MS	CH	SAL	TR	STX	MTX	CTX
t_1	Mark	Ballin	M	304	232-7667	Anthony	WA	25813	S	Y	70000	3	2000	0	2000
t_2	Chunho	Black	M	206	154-4816	Seattle	WA	98103	M	N	60000	4.63	0	0	0
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Conditional Functional Dependency

FDs are not sufficient to capture certain semantics of data. Conditional functional dependencies (CFDs), an extension of FDs, are capable of capturing FDs that hold partially on the data.

A CFD ϕ on R is a pair $(R: X \rightarrow Y, T_p)$, where:

- $X, Y \subseteq R$;
- $X \rightarrow Y$ is an FD, called *embedded FD* in the context of CFD; and
- T_p is called a *pattern tableau* of ϕ , where for every attribute $A \in X \cup Y$ and each pattern tuple $t_p \in T_p$, either $t_p[A]$ is a constant in the domain $\text{Dom}(A)$ of A , or $t_p[A]$ is a wild card '-'.

Conditional Function Dependency (Example)

- Consider the following sales table.
- The following FD does not hold:
 $\{\text{name, type, country}\} \rightarrow \{\text{price, tax}\}$
- For example, t_6 and t_7 are violating this FD.
- But the FD holds if (1) type is “clothing”; (2) country is “France”, type is “book” and tax is equal to 0; or (3) country is “UK”.
 - The third one is incorrect
- This constraint can be expressed by the CFD ($\{\text{name, type, country}\} \rightarrow \{\text{price, tax}, T_p\}$), with T_p as shown in the table on the bottom right of the slide.

TID	name	type	country	price	tax
t_1	Harry Potter	book	France	10	0
t_2	Harry Potter	book	France	10	0
t_3	Harry Potter	book	France	10	0.05
t_4	The Lord of the Rings	book	France	25	0
t_5	The Lord of the Rings	book	France	25	0
t_6	Algorithms	book	USA	30	0.04
t_7	Algorithms	book	USA	40	0.04
t_8	Armani suit	clothing	UK	500	0.05
t_9	Armani suit	clothing	UK	500	0.05
t_{10}	Armani slacks	clothing	UK	250	0.05
t_{11}	Armani slacks	clothing	UK	250	0.05
t_{12}	Prada shoes	clothing	USA	200	0.05
t_{13}	Prada shoes	clothing	USA	200	0.05
t_{14}	Prada shoes	clothing	France	500	0.05
t_{15}	Spiderman	DVD	UK	19	0
t_{16}	Star Wars	DVD	UK	29	0
t_{17}	Star Wars	DVD	UK	25	0
t_{18}	Terminator	DVD	France	25	0.08
t_{19}	Terminator	DVD	France	25	0
t_{20}	Terminator	DVD	France	20	0

name	type	country	price	tax
-	clothing	-	-	-
-	book	France	-	0
		UK		

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- Consider once again our Tax table examples.
- Can the following constraint be expressed using a functional dependency or a conditional functional dependency:
 - if two persons live in the same state, the one with lower salary has a lower tax rate
 - We cannot

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Denial Constraints

- As powerful as CFDs are, they are still not capable of capturing many real life data quality rules, such as:
 - if two persons live in the same state, the one with lower salary has a lower tax rate
- Denial constraints (DCs), a universally quantified first order logic formalism, which subsume FDs and CFDs

Denial Constraints (DCs)

Formal Definition:

$$\varphi : \forall t_\alpha, t_\beta, t_\gamma, \dots \in R, \neg(P_1 \wedge \dots \wedge P_m)$$

$$P_i: t_x.A \theta t_y.B \text{ or } t_x.A \theta c$$

$$x, y \in \{\alpha, \beta, \dots\}, \text{ and } A, B \in R, c \text{ is a constant}$$

- A universal constraint dictates a set of predicate cannot be true together
- Each predicate express a relationship between two cells, or a cell and a constant

Denial Constraints: Examples

Specify the following rules using denial constraints:

- The functional dependency: $\text{ZIP} \rightarrow \text{State}$
- Two people cannot have the same phone number
- If a person lives in the city of “Los Angeles”, then he is definitely in the state of California “CA”
- If two persons live in the same state, the one with lower salary has a lower tax rate

Denial Constraints: Examples

Specify the following rules using denial constraints:

- The functional dependency: $ZIP \rightarrow State$

$$\forall t_\alpha, t_\beta \in R, \neg(t_\alpha.ZIP = t_\beta.ZIP \wedge t_\alpha.ST \neq t_\beta.ST)$$

- Two people cannot have the same phone number
- If a person lives in the city of “Los Angeles”, then he is definitely in the state of California “CA”
- If two persons live in the same state, the one with lower salary has a lower tax rate

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- Two people cannot have the same phone number

$$\forall t_\alpha, t_\beta \in R, \neg(t_\alpha.AC = t_\beta.AC \wedge t_\alpha.PH = t_\beta.PH)$$

- If a person lives in the city of “Los Angeles”, then he is definitely in the state of California “CA”
- If two persons live in the same state, the one with lower salary has a lower tax rate

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- Two people cannot have the same phone number

$$\forall t_\alpha, t_\beta \in R, \neg(t_\alpha.AC = t_\beta.AC \wedge t_\alpha.PH = t_\beta.PH)$$

- If a person lives in the city of “Los Angeles”, then he is definitely in the state of California “CA”

$$\forall t_\alpha \in R, \neg(t_\alpha.CT = \text{'Los Angeles'} \wedge t_\alpha.ST \neq \text{'CA'})$$

- If two persons live in the same state, the one with lower salary has a lower tax rate

Functional Dependency (Example)

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Denial Constraints: Examples

Specify the following rules using denial constraints:

- The functional dependency: $ZIP \rightarrow State$

$$\forall t_\alpha, t_\beta \in R, \neg(t_\alpha.ZIP = t_\beta.ZIP \wedge t_\alpha.ST \neq t_\beta.ST)$$

- Two people cannot have the same phone number

$$\forall t_\alpha, t_\beta \in R, \neg(t_\alpha.AC = t_\beta.AC \wedge t_\alpha.PH = t_\beta.PH)$$

- If a person lives in the city of “Los Angeles”, then he is definitely in the state of California “CA”

$$\forall t_\alpha \in R, \neg(t_\alpha.CT = \text{'Los Angeles'} \wedge t_\alpha.ST \neq \text{'CA'})$$

- If two persons live in the same state, the one with lower salary has a lower tax rate

$$\forall t_\alpha, t_\beta \in R, \neg(t_\alpha.ST = t_\beta.ST \wedge t_\alpha.SAL < t_\beta.SAL \wedge t_\alpha.TR > t_\beta.TR)$$

Inclusion Dependency

- An inclusion dependency ϕ for two relations $(R1, R2)$ is defined as $R1[X] \subseteq R2[Y]$, where (X, Y) are lists of attributes in $(R1, R2)$, and $|X| = |Y|$.
- An instance pair (I_1, I_2) satisfies IND ϕ if for any tuple $t_\alpha \in I_1$, there exists a tuple $t_\beta \in I_2$, such that $t_\alpha[X] = t_\beta[Y]$.
- In other words, for any $t_\alpha \in I_1$, if there is no tuple $t_\beta \in I_2$ that satisfies $t_\alpha[X] = t_\beta[Y]$, then either $t_\alpha \in I_1$ is incorrect, or there is a tuple missing in $R2$.
- Inclusion dependencies may also refer to only one relation, i.e., $R1$ is the same as $R2$.

Inclusion Dependency: Example

- Consider the relation about tax records presented previously as Relation R_1 , and a second employee relation $R_2(\text{EID}, \text{FirstName}, \text{LastName})$ that keeps all the employee IDs, and their first and last names.

- A valid IND would be

$$R_1[\text{FN}, \text{LN}] \subseteq R_2[\text{FirstName}, \text{LastName}]$$

which means that if there is a tax record about a person, then that person must appear in the employee table.

Matching Dependencies (MDs)

Tran	FN	LN	St	City	AC	Post	Phn	Item
	Robert	Brady	5 Wren St	Ldn	020	WC1H 9SE	3887644	watch
	Robert	Brady	Null	Ldn	020	WC1E 7HX	3887644	necklace

Master: Card

FN	LN	St	City	AC	Zip	Tel
Robert	Brady	5 Wren St	Ldn	020	WC1H 9SE	3887644

MD: $\text{Tran}[\text{LN}, \text{City}, \text{ST}, \text{Post}] = \text{card}[\text{LN}, \text{City}, \text{St}, \text{Zip}] \Rightarrow \text{Tran}[\text{FN}, \text{Phn}] = \text{Card}[\text{FN}, \text{Tel}]$

Other Kinds of Dependencies

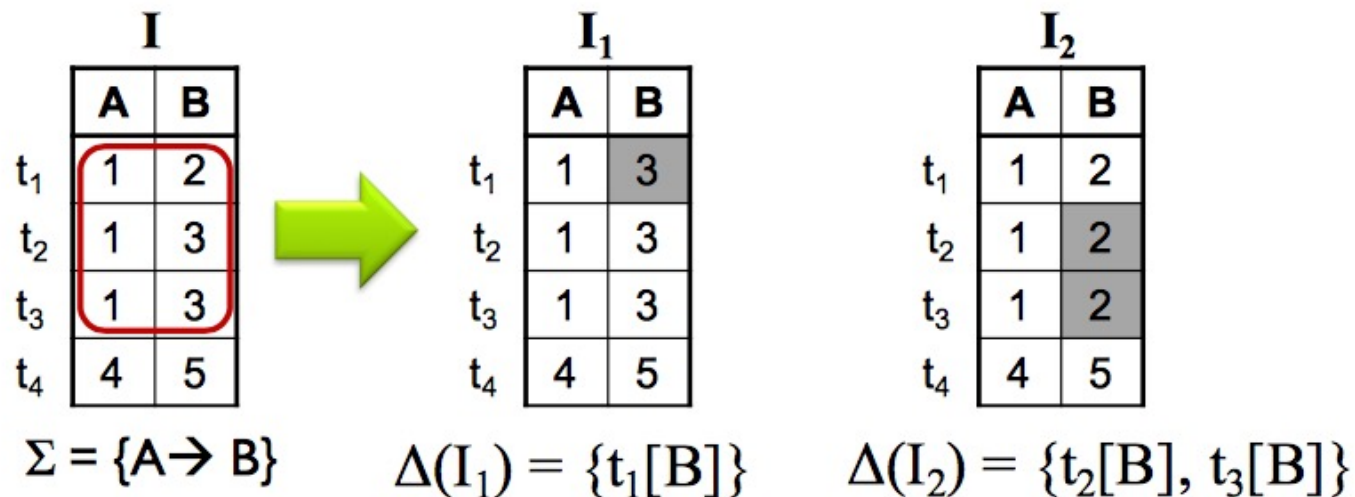
- Numerical Functional Dependencies
- Editing Rules
- Sherlock Rules

How to automate Data Repair?

- Most automatic repairing techniques adopt the “minimality” of repairs principle
- Repairing techniques in practice are manual and semi-automatic at best

Data repairs by Value Update

- I is a **dirty** database if $I \not\models \Sigma$, and I_j is a **repair** for I if $I_j \models \Sigma$
- For a repair I_j , $\Delta(I_j)$ is the set of changed cells in I_j



Data repairs by Value Updates

- Cardinality-Minimal repairs

- Commonly used in obtaining a single repair automatically
- Repairs with the minimum number of changes
- I_1 is Card-Min iff $\nexists I_2$ s.t. $|\Delta(I_2)| < |\Delta(I_1)|$

FD: {A → B}

	A	B
t ₁	1	2
t ₂	1	3
t ₃	1	3
t ₄	4	5

→

I ₁	
A	B
1	3
1	3
1	3
4	5

I ₂	
A	B
1	2
1	2
1	2
4	5

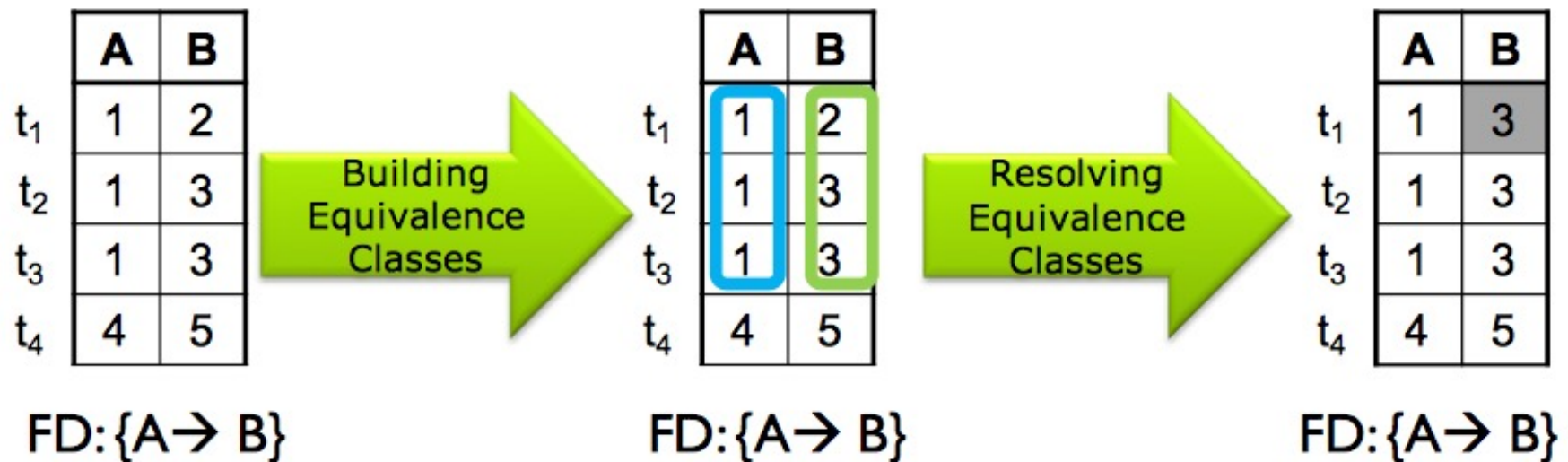
I ₃	
A	B
1	5
1	5
1	5
4	5

I ₄	
A	B
7	3
1	3
1	3
4	5

FD: {A → B} is satisfied by I₁ (green checkmark), but not by I₂, I₃, or I₄ (red X).

2. CONSTRAINT-BASED DATA REPAIRING

FD-based repairing



Repair Using Denial Constraints

- We will discuss the case of holistic data cleaning [1].
- Holistic data cleaning automatically detects violations of multiple Integrity constraints (ICs), and captures the cells that are more likely to be erroneous through the overlapping violations.
- Two violations are overlapping if they involve at least one common cell.
- We illustrate this using an example.

[1] <https://cs.uwaterloo.ca/~ilyas/papers/XuICDE2013.pdf>

Holistic Data Cleaning: Example

- Consider the following table.
- Every tuple specifies an employee in a company with her identification(TID), name (FN, LN), role (ROLE), ZIP (ZIP), state (ST), and salary (SAL).
- Consider two data quality rules.
- The first is a functional dependency (FD) stating that ZIP determines ST: $ZIP \rightarrow State$
- The second rule specifies that for two employees in the same state, the one whose role is manager cannot earn less than the one whose role is employee.
 - Is this a functional dependency or a conditional functional dependency?

TID	FN	LN	ROLE	ZIP	ST	SAL
t_1	Anne	Nash	E	85376	NY	110
t_2	Mark	White	M	90012	NY	80
t_3	Mark	Lee	E	85376	AZ	75

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- The first is a functional dependency (FD) stating that ZIP determines ST: $ZIP \rightarrow State$
- The second rule specifies that for two employees in the same state, the one whose role is manager cannot earn less than the one whose role is employee.
 - Is this a functional dependency or a conditional functional dependency?
 - Neither.
 - How can we specify it?

TID	FN	LN	ROLE	ZIP	ST	SAL
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- The second rule specifies that for two employees in the same state, the one whose role is manager cannot earn less than the one whose role is employee.
 - Is this a functional dependency or a conditional functional dependency?
 - Neither.
 - It can be specified as a Denial Constraint.

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- Every tuple specifies an employee in a company with her identification(TID), name (FN, LN), role (ROLE), ZIP (ZIP), state (ST), and salary (SAL).
- Consider two data quality rules.
- The first is a functional dependency (FD) stating that ZIP determines ST: $ZIP \rightarrow State$
- The second rule specifies that for two employees in the same state, the one whose role is manager cannot earn less than the one whose role is employee.
- *What are the cells that you think are likely to be erroneous if we examine both e_1 and e_2 ?*

TID	FN	LN	ROLE	ZIP	ST	SAL
t_1	Anne	Nash	E	85376	NY	110
t_2	Mark	White	M	90012	NY	80
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- Given the first rule, we can see that a set e_1 of four cells $\{t_1[ZIP], t_1[ST], t_3[ZIP], t_3[ST]\}$ in t_1 and t_3 present a violation for this FD: they have the same value for the city, but different states.
- Given the second rule, a set e_2 of six cells $\{t_1[ROLE], t_1[ST], t_1[SAL], t_2[ROLE], t_2[ST], t_2[SAL]\}$ in t_1 and t_2 are violating the rule, since employee White, who is a manager, is earning less than Nash, who is an employee.
- Looking at e_1 individually, one cannot tell which one of four cells in e_1 is wrong; similarly, looking at e_2 alone is also not sufficient.

Holistic Data Cleaning: Example

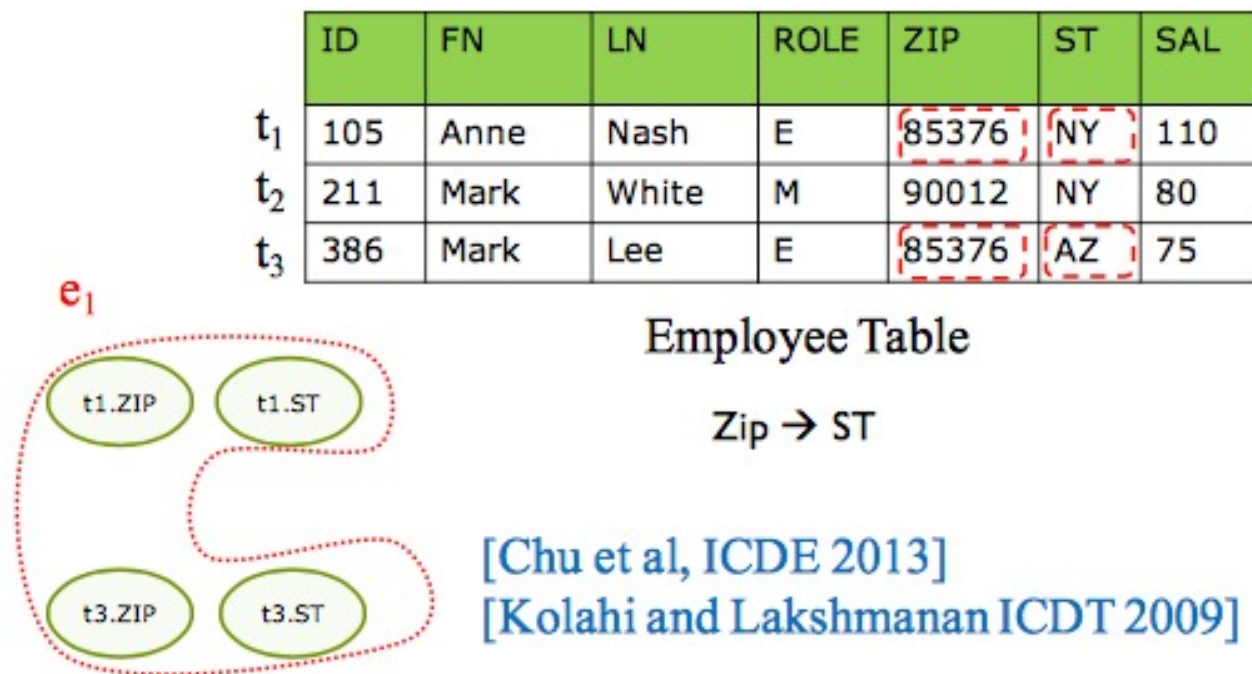
- Consider the following table.
- Every tuple specifies an employee in a company with her identification(TID), name (FN, LN), role (ROLE), ZIP (ZIP), state (ST), and salary (SAL).
- Consider two data quality rules.
- The first is a functional dependency (FD) stating that ZIP determines ST: $ZIP \rightarrow State$
- The second rule specifies that for two employees in the same state, the one whose role is manager cannot earn less than the one whose role is employee.
- Looking at both e_1 and e_2 , detect that $t_1[ST]$ is likely to be wrong since it participates in the two sets e_1 and e_2 .

TID	FN	LN	ROLE	ZIP	ST	SAL
t_1	Anne	Nash	E	85376	NY	110
t_2	Mark	White	M	90012	NY	80
t_3	Mark	Lee	E	85376	AZ	75

- Given the first rule, we can see that a set e_1 of four cells $\{t_1[ZIP], t_1[ST], t_3[ZIP], t_3[ST]\}$ in t_1 and t_3 present a violation for this FD: they have the same value for the city, but different states.
- Given the second rule, a set e_2 of six cells $\{t_1[ROLE], t_1[ST], t_1[SAL], t_2[ROLE], t_2[ST], t_2[SAL]\}$ in t_1 and t_2 are violating the rule, since employee White, who is a manager, is earning less than Nash, who is an employee.
- Looking at e_1 individually, one cannot tell which one of four cells in e_1 is wrong; similarly, looking at e_2 alone is also not sufficient.

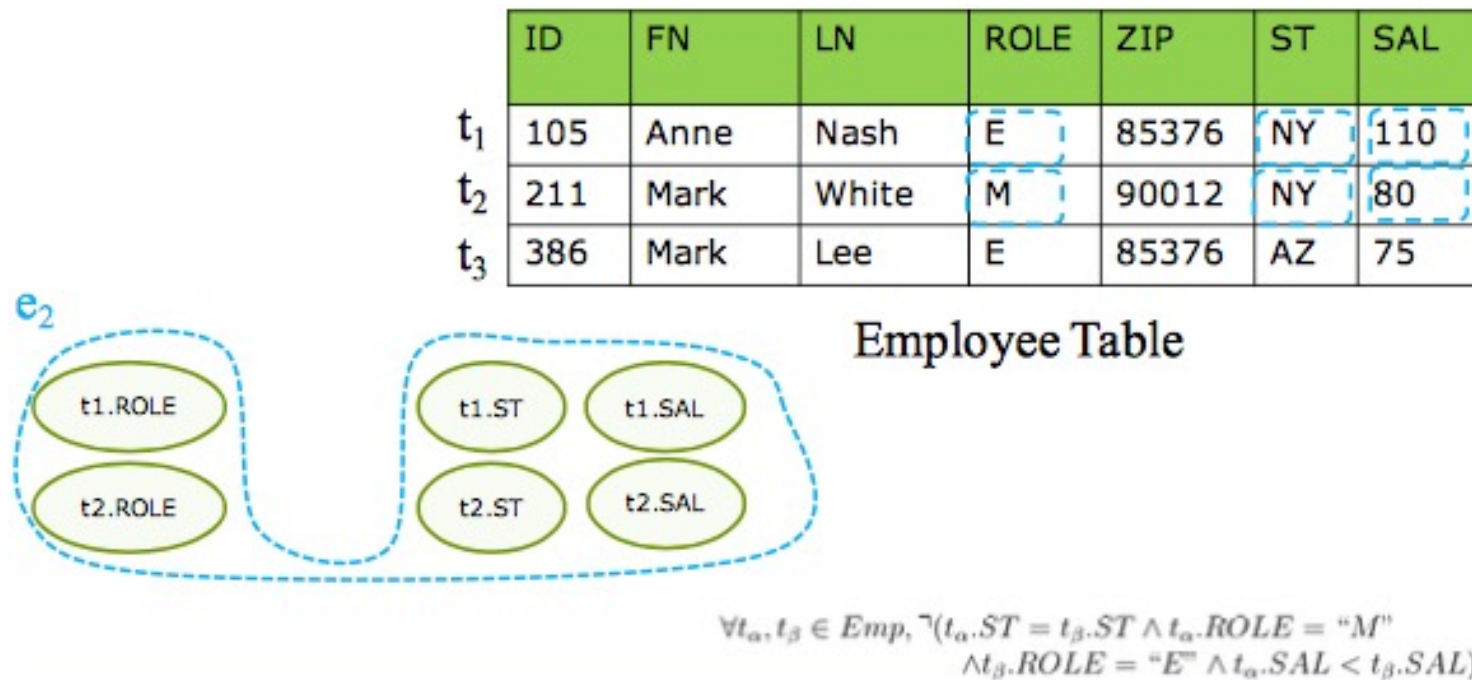
Holistic Constraint-based Repairing

Hypergraph-based approach: Each cell in the DB is a vertex, each set of tuples violating a constraint form a hyperedge



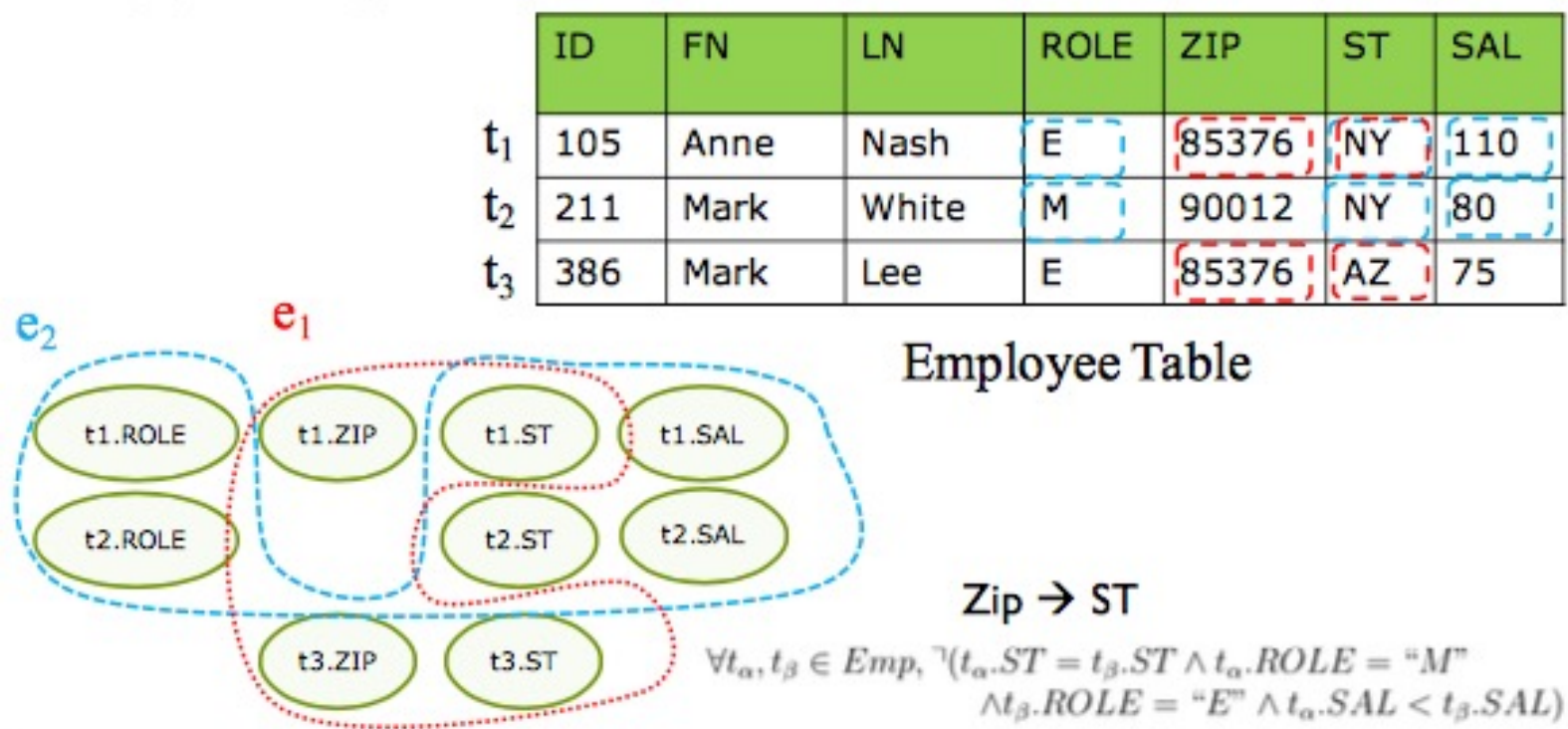
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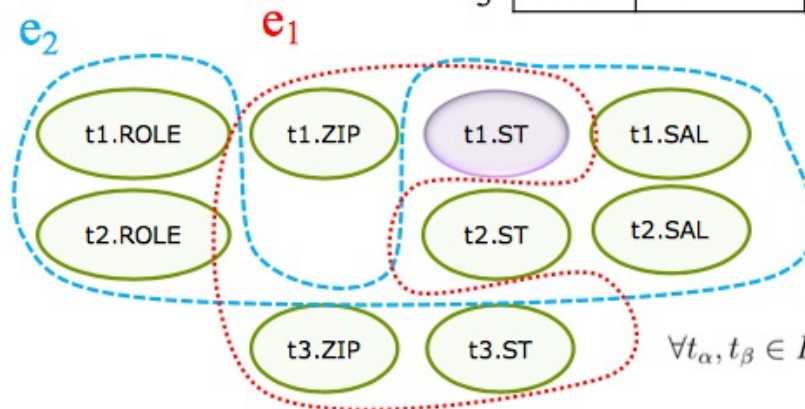


Step1: Minimal Vertex Cover

- A minimal set of vertices that are intersecting with every hyperedge

	ID	FN	LN	ROLE	ZIP	ST	SAL
t_1	105	Anne	Nash	E	85376	NY	110
t_2	211	Mark	White	M	90012	NY	80
t_3	386	Mark	Lee	E	85376	AZ	75

Employee Table



Zip $\rightarrow$ ST

$$\forall t_\alpha, t_\beta \in Emp, \neg(t_\alpha.ST = t_\beta.ST \wedge t_\alpha.ROLE = "M" \wedge t_\beta.ROLE = "E" \wedge t_\alpha.SAL < t_\beta.SAL)$$

Step2: Collect Repair Requirements

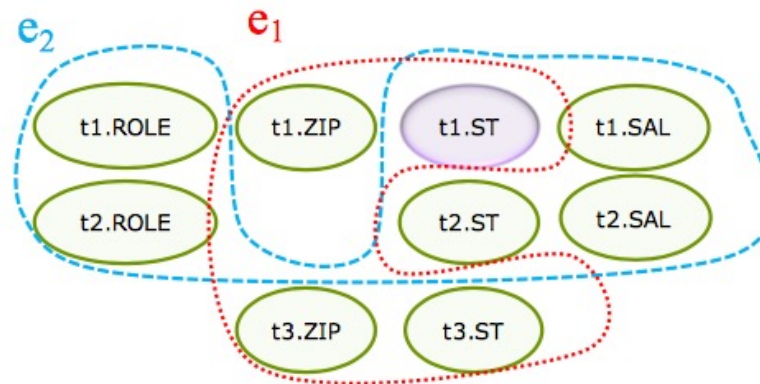
- A set of conditions that need to be satisfied to resolve all violations

Condition to resolve e_1 by changing $t1.ST$:

$$t1.ST = t3.ST$$

Condition to resolve e_2 by changing $t1.ST$:

$$t1.ST \neq t2.ST$$



Zip $\rightarrow$ ST

$$\forall t_\alpha, t_\beta \in Emp, \neg(t_\alpha.ST = t_\beta.ST \wedge t_\alpha.ROLE = "M" \wedge t_\beta.ROLE = "E" \wedge t_\alpha.SAL < t_\beta.SAL)$$

Step3: Get Updates

- A set of assignments satisfying the conditions, with minimal number of changed cells

$t_1.ST = t_3.ST$

$t_1.ST \neq t_2.ST$

	ID	FN	LN	ROLE	ZIP	ST	SAL
t_1	105	Anne	Nash	E	85376	NY	110
t_2	211	Mark	White	M	90012	NY	80
t_3	386	Mark	Lee	E	85376	AZ	75

Gradually increase the number of cells that are going to be changed, until reach a solution

Suppose we only want to change $t_1.ST$

$t_2.ST = NY$

$t_3.ST = AZ$

AZ

Exercise

Student ID	Semester	Lecture	TA
1234	5	Numerical Methods	John
1221	4	Numerical Methods	Smith
1234	6	Visual Computing	Bob
1201	2	Big Data	Peter
1201	2	Physics II	Simon

- Functional dependencies
 - StudentID $\rightarrow$ Semester
 - Lecture $\rightarrow$ TA
- Detect the cells that may be erroneous
- Identify the cells that needs to be modified, and their new values

Exercise

Student ID	Semester	Lecture	TA
1234	5	Numerical Methods	John
1221	4	Numerical Methods	Smith
1234	6	Visual Computing	Bob
1201	2	Big Data	Peter
1201	2	Physics II	Simon

- Functional dependencies
 - StudentID $\rightarrow$ Semester
 - Lecture $\rightarrow$ TA
- **Detect the cells that may be erroneous**
- Identify the cells that needs to be modified, and their new values

Exercise

Student ID	Semester	Lecture	TA
1234	5	Numerical Methods	John
1221	4	Numerical Methods	Smith
1234	6	Visual Computing	Bob
1201	2	Big Data	Peter
1201	2	Physics II	Simon

- Functional dependencies
 - $\text{StudentID} \rightarrow \text{Semester}$
 - $\text{Lecture} \rightarrow \text{TA}$
- Detect the cells that may be erroneous
- **Identify the cells that needs to be modified, and their new values**
 - There are multiple possible reopairs that require a minimal number of cell modifications, namely two

Student ID	Semester	Lecture	TA
1234	5	Numerical Methods	John
1221	4	Numerical Methods	Smith
1234	6	Visual Computing	Bob
1201	2	Numerical Methods	Peter
1201	2	Physics II	Simon

Student ID	Semester	Lecture	TA
1234	5	Numerical Methods	John
1221	4	Numerical Methods	John
1234	5	Visual Computing	Bob
1201	2	Big Data	Peter
1201	2	Physics II	Simon

Student ID	Semester	Lecture	TA
1235	5	Numerical Methods	Smith
1221	4	Numerical Methods	Smith
1234	6	Visual Computing	Bob
1201	2	Big Data	Peter
1201	2	Physics II	Simon

These are not the only repairs with minimal changes